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Upper Extremity Study Guide MSK

Region	Brief Description of Exam	Positive Test Findings	What diagnosis is a positive test consistent with?
Neck Special Tests			
Spurling Test  	Have patient extend, sidebend and rotate head to side being tested (to narrow the cervical intervertebral foramen). If no radicular symptoms, place axial load gently (5-10 lbs) from top of head to reproduce symptoms of paresthesia. Apply axial load up to 5 seconds.	Reproduction of paresthesia (numbness, tingling symptoms) in neck or arm. If there is a reproduction of symptoms with set up and no axial force, no need to apply axial force.	Cervical radiculopathy
Lhermitte Sign 	Instruct patient to flex head forward, chin to chest and hold for up to 30 seconds. Forward flexion of cervical spine is thought to put pressure on dorsal column of spinal cord.	Reproduction of paresthesia down spine or back	Spinal cord compression in cervical region (myelopathy, MS, spondylosis)
Adson Test 	Position patient sitting. Extend and rotate head to the affected side. *Extend* arm on the same side and assess radial pulse. Have patient take a deep breath and hold for 10-15 seconds. These actions narrow thoracic outlet structures (scalenes, pec minor, costoclavicular space) and increase compression on brachial plexus or brachial artery/vein.	Reproduction of paresthesia and decreased radial pulse.	Thoracic outlet syndrome
Roos Test (elevated arm stress test) 	Have patient hold both arms up (abducted at shoulders, elbows flexed 90 degrees, externally rotated). Have patient open and close hands for 3 minutes. Stresses structures of thoracic outlet. (scalenes, pec minor, costoclavicular space)	Reproduction of paresthesias, cramping of hand(s) due to lower blood flow. Caution: High false positive due to fatigue	Thoracic outlet syndrome

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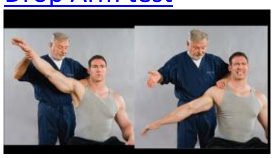
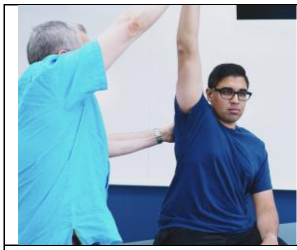
Shoulder Exam (Intrinsic tests for Rotator Cuff)			
Empty Can test 	Have patient abduct at shoulder to 90 degrees, bring arms forward 30 degrees in horizontal plane, and internally rotate arms so thumbs point down to floor. Examiner pushes inferiorly at distal end of humerus and patient resists force (holds arms up)	Reproduction of pain at shoulder- supraspinatus tendonitis, partial tear. Weakness with resistance - possible supraspinatus tear/rupture.	Supraspinatus tendonitis or rupture
Drop Arm test 	Assist patient to abduct both arms or one arm at a time to 90 degrees and have patient lower slowly. A seamless symmetrical descent is a normal exam.	A fast drop is suggestive of tendonitis or partial tear. A sudden or uncontrolled drop is suggestive of full tear.	Pathology of supraspinatus (0-15 degrees) or deltoid (15-90 degrees)
Infraspinatus/Teres Minor Isolated resistance 	Have patient place arms at their side and flex elbows to 90 degrees with thumbs up. Provide resistance as patient presses forearms outward (external rotation)	Weakness with resistance (external rotation)	Infraspinatus and/or Teres Minor tendonitis or tear
Hornblower Test Teres minor 	The patient is seated or standing. The examiner places the patient's arm to 90°in the scapular plane and flexes the elbow to 90°. The patient is then asked to externally rotate against resistance. The test is positive if the patient is unable to perform external rotation.	Reproduction of pain Weakness with resistance (will not be tested on practical)	To test for teres minor tear.
Lift off Test Subscapularis 	Have patient place hand behind their back with palm facing away from body. Instruct patient to "lift off" their hand (tests internal rotation) as they press their palm away from their back against your resistance.	Weakness with resistance partial tear or rupture (internal rotation)	Subscapularis tendonitis or tear
Subacromial Impingement Tests			
Neer's Sign	Place your hand on the posterior scapula for stabilization. With your other arm, hold your patient's arm (with internal rotation at shoulder) at the wrist and bring shoulder into full flexion.	Reproduction of patient's pain in shoulder	Rotator cuff tendonitis or possible tear, subacromial impingement

	This maneuver compresses the greater tuberosity of humerus into anterior acromion, narrowing the subacromial space		
Hawkins Impingement sign 	Flex patient's shoulder to 90°, flex elbow to 90°, place forearm in neutral position. Passively internally rotate at shoulder (compresses space underneath acromion)	Reproduction of patient's pain in shoulder	Rotator cuff tear or impingement syndrome/rotator cuff tendonitis.

Extrinsic Shoulder Tests structures other than rotator cuff			
Crossover test 	With Arm flexed 90° move arm across body in adduction. Compresses AC joint.	Reproduction of pain at AC joint	AC joint pathology (arthritis)
Shoulder Apprehension test 	Supine or sitting. Place patient's arm in 90° abduction, then maximal external rotation (to stress anterior joint capsule of shoulder). If patient demonstrates apprehension that shoulder will dislocate, this is positive for anterior glenohumeral joint instability. Second phase to confirm: Fowlers Relocation test, with hand on anterior shoulder provide firm pressure. This relieves sense of apprehension.	+ apprehension is positive sign. Relief with anterior pressure (Fowlers) is confirmatory.	Anterior glenohumeral joint instability (chronic) *(Not done in acute dislocations)*
O'Brien's Active Compression Test 	Two steps: Position arm in 90° flexion and *20-30° horizontal adduction* (this focuses more on labrum) 1) Position the shoulder in full internal rotation (thumb down) and ask patient to resist you. 2) Second part: have patient externally rotate (rotate thumb up) and resist.	Positive test is when pain is elicited with thumb down and pain relieved with the thumb up.	Pain at top of shoulder- AC joint pathology, Pain in Shoulder- SLAP lesion

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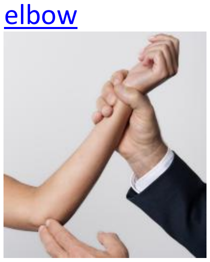
Speed's Test 	While stabilizing at the shoulder, place patient's arm in flexion, elbow extended and supinated. Examiner pushes inferiorly and patient resists downward force.	Pain in the bicipital groove suggests biceps tendonitis or SLAP lesion. Weakness - a severe biceps tendon strain may be present.	Biceps tendonitis or SLAP lesion.
Yergason's Test 	Have patient flex elbow, forearm pronated with arm next to torso. While palpating the bicipital groove, ask the patient to supinate against your resistance.	A positive test: feeling the biceps tendon sublux out of groove (transverse humeral ligament is torn) or if pain, then suspect biceps tendinosis or SLAP lesion.	Biceps tendon pathology or SLAP lesion
Elbow special tests			
Cozen's Test 	With one hand, palpate just distal to lateral epicondyle, and support elbow in extension. Then, ask your patient to make a fist in pronation, extend their wrist and radially deviate their wrist (turn towards thumb) Ask patient hold this position while examiner applies resistance.	Positive if patient reports pain in the lateral epicondyle with resistance.	Lateral Epicondylitis
Reverse Cozen 	With one hand, palpate just distal to medial epicondyle, and support elbow in extension. Then, ask your patient to make a fist in supination, flex their wrist and ulnar deviate their wrist (turn towards 5th digit) Ask patient hold this position while examiner applies resistance.	Positive if patient reports pain in the medial epicondyle with resistance.	Medial Epicondylitis
Tinel's Test of the elbow 	Using fingers, tap repetitively over ulnar nerve where it runs posteriorly to medial epicondyle.	A positive test replicates sensation of numbness/tingling in ring and pinky.	Cubital tunnel syndrome (ulnar nerve compression)
Wrist/Hand Tests			

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Tinel's sign of the wrist 	Supinate the patient's forearm and extend the wrist. Tap distal of the flexor retinaculum where median nerve exits wrist.	Positive test is reproduction of paresthesia's of the first three fingers.	Carpal tunnel syndrome (median nerve compression)
Phalen's test 	Have the patient press dorsum of hands together and hold wrists in flexion for 60 seconds.	A positive test is reproduction of paresthesia in first three fingers.	Carpal Tunnel Syndrome (median nerve compression)
Carpal Compression Test 	Patient position in standing or sitting. The patient forearm is supinated and then the examiner applies direct pressure over the carpal tunnel (median nerve) between the thenar and hypothenar eminence for 30 seconds.	A positive test is reproduction of paresthesia in first three fingers.	Carpal Tunnel Syndrome (median nerve compression)
Finkelstein's test 	Have the patient flex their thumb and clench fist over thumb. Have patient turn wrist into ulnar deviation. Stretches the tendon sheath of extensor pollicus brevis, abductor pollicus longus)	Positive test is pain at the radial aspect of thumb and wrist.	DeQuervain's Tenosynovitis